

BRAC University
Department of Electrical and Electronic Engineering
EEE-103IL

Lab Assignment — Experiment 3B: Loops

General Instructions

- Solve every task in **C**.
- All inputs must be read from the keyboard with `scanf` — do not hard-code the input values inside the program.
- You may use only the constructs covered so far: variables, arithmetic, `if--else`, and loops (`for`, `while`, `do-while`). **Library functions such as `pow()` are not allowed.**
- Every pattern must be produced by **nested loops**. A program that simply prints the pattern with a series of fixed `printf` lines will receive **zero**.
- Your output should match the format of the sample runs shown with each task.

Task 1: Even Sum and Odd Sum

Read a positive integer N . Compute the sum of all even numbers and the sum of all odd numbers in that range *separately* and print both.

```
Sample Input:
10
Sample Output:
Sum of even numbers = 30
Sum of odd numbers  = 25
```

Task 2: Digits Smaller than 5

Read a positive integer N (it may have any number of digits). Using a loop with the operators `%` and `/`, determine:

- how many digits N has in total, and
- how many of those digits are **smaller than 5**.

Hint: $N \% 10$ gives the last digit of N and $N = N / 10$ removes it.

```
Sample Input:
8241937
Sample Output:
Total digits      = 7
Digits smaller than 5 = 4
```

(In the example, the digits smaller than 5 are 2, 4, 1, 3.)

Task 3: Exponential Series

The exponential function can be approximated by the truncated series

$$e^x \approx \sum_{k=0}^N \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^N}{N!}.$$

Read a real number x and an integer N , then compute this sum. For **every** term of the outer loop you must compute:

- x^k with an **inner loop** (repeated multiplication — `pow()` is not allowed) and
- $k!$ with another **inner loop** (repeated multiplication).

Finally, print the result.

```
Sample Input:
2 10
Sample Output:
Series approximation = 7.388995
```

Task 4: Sum of Self-Powers

Read an integer N ($1 \leq N \leq 9$) and compute

$$S = 1^1 + 2^2 + 3^3 + \dots + N^N,$$

where each power k^k must be computed with an **inner loop** of repeated multiplication. Print every term and the running sum as the loop progresses, then the final value of S .

```
Sample Input:
4
Sample Output:
k = 1 : term = 1      sum = 1
k = 2 : term = 4      sum = 5
k = 3 : term = 27     sum = 32
k = 4 : term = 256    sum = 288
Final sum S = 288
```

Task 5: Inverted Letter Triangle

Read an integer n ($2 \leq n \leq 26$) and print an inverted triangle of the first n capital letters: the first row shows the letters A up to the n -th letter, and each following row drops the last letter.

Hint: in C, characters are numbers — `65 + 2` is `'C'`. Print with `%c`.

```
Sample Input:
5
Sample Output:
A B C D E
A B C D
A B C
A B
A
```

Task 6: Hollow Square and Rhombus

Read an integer n ($n \geq 3$) and print the two shapes below, one after the other.

- A **square with a blank middle** (stars only on the border, spaces inside).
- A **rhombus**: n rows of n stars, where each row is shifted one space further left than the row above it (the first row is shifted the most).

Hint for (a): print a star only when the row or the column is the first or the last one; otherwise print a space.

Sample Input:

5

Sample Output:

* *

* *

* *

Task 7: Diamond

Read an integer n and print a **diamond** of stars whose upper half has n rows: row i of the upper half contains $2i - 1$ stars centred with leading spaces, and the lower half mirrors the upper half.

Sample Input:

4

Sample Output:

 *

 *